Egg consumption and the risk of type 2 diabetes mellitus: a case-control study.

OBJECTIVE: Type 2 diabetes mellitus appears to involve an interaction between susceptible genetic backgrounds and environmental factors including highly calorific diets. As it is important to identify modifiable risk factors that may help reduce the risk of type 2 diabetes mellitus, the aim of the present study was to determine the association between egg consumption and the risk of type 2 diabetes mellitus. DESIGN: A specifically designed questionnaire was used to collect information on possible risk factors of type 2 diabetes mellitus. The odds ratios and 95 % confidence intervals for type 2 diabetes mellitus were calculated by conditional logistic regression. SETTING: A case-control study in a Lithuanian out-patient clinic was performed in 2001. SUBJECTS: A total of 234 cases with a newly confirmed diagnosis of type 2 diabetes mellitus and 468 controls free of the disease. RESULTS: Variables such as BMI, family history of diabetes, cigarette smoking, education, morning exercise and plasma TAG level were retained in multivariate logistic regression models as confounders because their inclusion changed the value of the odds ratio by more than 10 % in any exposure category. After adjustment for possible confounders more than twofold increased risk of type 2 diabetes mellitus was determined for individuals consuming 3-4·9 eggs/week (OR = 2·60; 95 % CI 1·34, 5·08) and threefold increased risk of the disease was determined for individuals consuming ≥5 eggs/week (OR = 3·02; 95 % CI 1·14, 7·98) compared with those eating <1 egg/week. CONCLUSIONS: Our data support a possible relationship of egg consumption and increased risk of type 2 diabetes mellitus.